This problem set is due *by 11:59pm* on **Wednesday, February 26**. It should be submitted through Canvas. All submissions should be typeset using Latex.

Problem 1 (Lyapunov Analysis and Optimal Control) *Consider the following nonlinear system:*

$$\dot{x} = x^3 + xu \tag{1}$$

Now consider the scalar function

$$V(x) = 2x^2 \tag{2}$$

and the controller

$$u(x) = -x\frac{\partial V(x)}{\partial x}. \tag{3}$$

a) *(**3 points**) Analyze the stability about $x = 0$ in the sense of Lyapunov. Is it stable? Asymptotically stable? Exponentially stable? Show your work.*

b) *(**4 points**) Find a cost function for which this controller is optimal.*

Problem 2 (Regions of Attraction) *Submit all code for this problem with the filename "vanderpol_roa.m" The reverse time Van der Pol oscillator is frequently used to test region of attraction algorithms. The equations of motion for this system are:*

$$\dot{x_1} = -x_2 \tag{4}$$

$$\dot{x_2} = x_1 - (1 - x_1^2)x_2 \tag{5}$$

a) *(**2 points**) Plot the vector field for this system. In a few sentences, provide an intuitive understanding of the system dynamics. You may find the file "example_vector_field.m" helpful.*

b) *(**2 points**) The Control System Toolbox provides a function (lyap) which allows you to find a valid Lyapunov function for a linear system of the form $V(x) = x^T P x < \rho$. Linearize the equations for the Van der Pol oscillator about the origin and use MATLAB to find a valid P. Note, the lyap function solves the equation $AP + PA^T + Q = 0$, which is the transpose of the form we use in class.*

c) *(**4 points**) Now that we have a valid Lyapunov function for the linear system, we can use it to find a valid region of attraction. Using $V < \rho$, derive $\dot{V}$ for the nonlinear system. Now, using this $\dot{V}$, write a grid based algorithm to search over the two dimensional state space to find the largest ρ such that $\dot{V} < 0$. Plot your resulting ellipse on your vector plot. You may find the MATLAB script "example-level-set.m" helpful for this task. It plots a level set for a function of the for $V = \mathbf{x}^T P \mathbf{x}$, where P is a positive definite matrix and $\mathbf{x}$ is a two dimensional vector.*

d) ***(5 points)*** *Now use SOS to find the largest ρ such that $\dot{V} < 0$. Plot the resulting ellipse on the vector field. For this part you will need to download two optimization packages for MATLAB. The first one, SeDuMi, is a semidefinite programming package. It is located at* `https://github.com/sqlp/sedumi`*. To install it, download the files and add the folders to the MATLAB path. The second package, SPOTless, can be found at* `https://github.com/mmt/spotless`*. Download the files, run* `spotinstall`*, and add the folders to the MATLAB path. You may find the MATLAB script "example-spot.m" helpful for this task.*

e) ***(2 points)*** *Explain any differences there may be between the ellipses you found in parts (c) and (d).*

f) ***(2 points)*** *Simulate some trajectories of this system from initial conditions which are inside and outside your estimated regions of attraction. Plot the results on your vector field.*

Problem 3 (Control Contraction Metrics) *In this problem, we will explore generating a global control contraction metric using sum-of-squares optimization. You will find the papers posted in CANVAS helpful for this problem. Consider the polynomial dynamical system*

$$\dot{x}_1 = -x_1 + x_3 \tag{6}$$

$$\dot{x}_2 = x_1^2 - x_2 - 2x_1x_3 + x_3 \tag{7}$$

$$\dot{x}_3 = -x_2 + u \tag{8}$$

a) ***(1 point)*** *Open the code "ccm_example" and complete the "gradients" function to compute an LQR controller linearized about the origin.*

b) ***(1 points)*** *In this problem, our candidate dual metric will be a matrix of polynomials described by*

$$\mathbf{W}(\mathbf{x}) = \mathbf{W}_0 + \mathbf{W}_1 x_1 + \mathbf{W}_2 x_1^2 + \mathbf{W}_3 x_2 + \mathbf{W}_4 x_2^2 + \mathbf{W}_5 x_1 x_2 \tag{9}$$

. The multiplier ρ will be given as $\rho(\mathbf{x}) = \rho_0 + \rho_1 x_1 + \rho_2 x_1^2$ Explain how this dual metric satisfies Condition 2 for the differential control input having the form $\delta\mathbf{u} = \mathbf{K}(\mathbf{x})\delta\mathbf{x}$. Show your work.

c) ***(2 points)*** *For this problem, we will be using different software to do sums-of-squares optimization than Problem Set 1. Like before, you will need to download two optimization packages for MATLAB. The first one, Mosek, is a semidefinite programming package. It is located at* `https://www.mosek.com/downloads/`*. Be sure to download the version for Ubuntu 18.04, even if you are running 20.04, to get MATLAB support. Add Mosek to your MATLAB path. The second package, YALMIP, can be found at* `https://yalmip.github.io/download/`*. The YALMIP documentation is quite extensive. The tutorial* `https://yalmip.github.io/`

`tutorial/sumofsquaresprogramming/` *will be quite helpful. Take some time to familiarize yourself with YALMIP. Using YALMIP variables, fill in the dual metric, the multiplier, and* $\dot{\mathbf{W}}$.

d) ***(3 points)*** *Fill in the control contraction condition P, which is referred to as Condition 1. Explain how you would show that a matrix of polynomials P is positive definite using SOS. HINT: the SOS constraint should be a scalar constraint.*

e) ***(5 points)*** *Set up the SOS constraints on P, W, and ρ and solve using YALMIP for the dual metric. Then, extract the numerical values from the YALMIP variables.*

f) ***(2 points)*** *Fill in the function "cost_function" to compute the geodesic, $\gamma(s)$, along which we integrate the differential feedback control. Also specify the boundary conditions for the optimization problem in the "computeGeodesic" method.*

g) ***(3 points)*** *Fill in the expression for the differential control in the function "evalDeltaK". Then fill in the code to integrate the differential control along the geodesic in "integrateDeltaK".*

h) ***(3 points)*** *Run the controller controller derived using the CCM and compare against the performance against LQR. Find a set of initial conditions for which the CCM-based controller can stabilize the origin but LQR can not. Submit plots of the resulting system trajectories and your code.*

Problem 4 (Lagrange Multipliers)

a) **(3 points)** *Using Lagrange multipliers, analytically find the values of x_1 and x_2 which minimize the cost function*

$$J(\mathbf{x}) = 2x_1^2 - 8x_1 - 4x_2 + 4x_2^2 + 12 \tag{10}$$

subject to the constraint

$$x_1 + x_2 = 2 \tag{11}$$

What is the value of this minimum cost?

b) **(3 points)** *Using MATLAB, submit a plot of the contours of this cost function, the linear constraint, and your optimal point in $x_1 - x_2$ plane.*

Problem 5 (Model-Based Policy Search) *In Lecture 7 and 8, we discussed trajectory optimization and model-based policy search. In this problem, we will explore different approaches for trajectory optimization for the cart-pole swing-up task.*

a) ***(2 points)*** *Describe the difference between direct and indirect trajectory optimization approaches. Describe the differences between shooting methods and collocation/transcription methods.*

b) ***(5 points)*** *Explore the code in* `cartpole_bptt.m` *and* `bptt.m`. *This code skeleton implements the Backpropagation Through Time algorithm (Adjoint Method). Define the cost matrices. Fill in the missing code in the* `compute_gradients` *so that we can apply the adjoint method (Backpropagation Through Time) to solve a swing-up trajectory for the cart-pole system. Here we assume our policy is parameterized with a parameter vector α, such that $\pi_\alpha(x_k)$. In this case, our policy is an open-loop policy such that $\alpha_k = u_k$. Submit your code and state-space plots of your swing-up trajectory.*

c) ***(5 points)*** *Explore the code in* `cartpole_dirtran.m` *and* `dirtran2.m`. *This code skeleton implements direct transcription. Define the cost matrices, time-step and number of time-steps. Assuming forward Euler integration for the dynamics constraint, fill in the evaluation of the cost and constraints in* `cost_function` *and* `constraints`. *Submit your code and state-space plots of your swing-up trajectory. Compare the results for direct transcription with Backpropagation Through Time. Discuss the reasons for any differences you observe.*